



Integrable semi-discretizations and self-adaptive moving mesh method for a generalized sine-Gordon equation

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Abstract

In the present paper, two integrable and one non-integrable semi-discrete analogues of a generalized sine-Gordon (sG) equation are constructed. The keys of the construction are the Bäcklund transformation of bilinear equations and appropriate definition of the discrete hodograph transformation. We construct N -soliton solutions for the semi-discrete analogues of the generalized sG equation in the determinant form. In the continuous limit, we show that the semi-discrete generalized sG equations converge to the continuous generalized sG equation. Furthermore, we propose four self-adaptive moving mesh methods for the generalized sG equation, two are integrable and two are non-integrable. Integrable and non-integrable self-adaptive moving mesh methods are proposed and used for simulations of regular, irregular and loop soliton while comparing with the Crank-Nicolson (C-N) scheme. The numerical solutions show that the proposed self-adaptive moving methods perform better than the C-N scheme.

Keywords Generalized sG equation · Integrable discretization · Discrete hodograph transformation · Self-adaptive moving mesh method

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1 Introduction

In this paper, we consider the integrable semi-discretizations of the following generalized sine-Gordon (sG) equation

$$u_{xt} = (1 + \nu \partial_x^2) \sin u, \quad (1.1)$$

where $u = u(x, t)$ is a scalar-valued function, ν is a real parameter, ∂_x^2 and the subscripts t and x appended to u denote partial differentiation. The generalized sG equation was derived for the first time in [1] using bi-Hamiltonian method. In the case $\nu = -1$, the integrability was shown with Lax pair and initial value problem of the equation was solved for decaying initial data [2]. The Riemann-Hilbert formalism was developed to obtain eigenfunctions of the Lax pair and construct traveling-wave solutions in [2]. Bilinear transformation method was used to construct multi-soliton solutions in the form of parametric representation and a variety of solutions which include kinks, loop solitons and breathers were obtained in [3]. Equation (1.1) with $\nu = 1$ was solved by Hirota's bilinear method [4]. It was shown that the equation exhibits kink and breather solutions and does not admit multi-valued solutions like loop solitons as obtained when $\nu = -1$.

It was shown that the generalized sG equation reduces to the short pulse (SP) equation in the short wave limit, and reduces to the sG equation in the long wave limit [3]. Therefore, we can say that the generalized sG equation is an interesting soliton equation lies between the SP equation and the sG equation. Due to the facts that the discrete sG equation has been well-known [5, 6], and the integrable discretizations of the SP equation was recently proposed in [7], it would be an interesting problem to look for the integrable discretization of the generalized sG equation. Integrable discretizations of soliton equations have received considerable attention recently [8–11]. Integrable semi- and fully discretizations of the SP equation [7], semi-discretization of (2+1)-dimensional Zakharov equation [12] were constructed via Hirota's bilinear method. Based on the compatibility between an integrable system and its Bäcklund transformation, a systematic procedure to derive discrete analogues of integrable PDEs via Hirota's bilinear method was proposed [13]. Most recently, an integrable discretization for a coupled SP equation [14, 15] was constructed [16].

In this paper, we attempt to construct integrable semi-discrete analogues of the generalized sG equation (1.1) with $\nu = -1$ by the same approach used in the Camassa-Holm equation [17] and SP equation [7], and apply them for the numerical simulations as integrable self-adaptive moving mesh (SAMM) method.

The paper is organized as follows. In Section 2, we review the bilinear equations and determinant solutions of the generalized sG equation with $\nu = -1$. In Section 3, we propose two integrable semi-discrete analogues, whose N -soliton solutions are also constructed in terms of determinant form. We then apply the two versions of the semi-discrete generalized sG equations as integrable SAMM method while comparing with a non-integrable SAMM method. Section 4 is devoted to conclusions and some discussions.

2 Bilinear equations and determinant solutions

In this section, we first review the results in [3] about the bilinear form of the generalized sG equation.

By introducing the new dependent variable r in accordance with the relation

$$r^2 = 1 + u_x^2, \quad (r > 0), \quad (2.1)$$

the generalized sG equation (1.1) with $\nu = -1$ is rewritten as

$$r_t + (r \cos u)_x = 0. \quad (2.2)$$

We then define the hodograph transformation $(x, t) \rightarrow (y, \tau)$ by

$$dy = r dx - r \cos u dt, \quad d\tau = dt, \quad (2.3)$$

The x and t derivatives are then rewritten in terms of the y and τ derivatives as

$$\frac{\partial}{\partial x} = r \frac{\partial}{\partial y}, \quad \frac{\partial}{\partial t} = \frac{\partial}{\partial \tau} - r \cos u \frac{\partial}{\partial y}. \quad (2.4)$$

With the new variables y and τ , (2.1) and (2.2) are recast into the form

$$r^2 = 1 + r^2 u_y^2, \quad (2.5)$$

$$\left(\frac{1}{r}\right)_\tau - (\cos u)_y = 0, \quad (2.6)$$

respectively. Further reduction is possible if one defines the variable ϕ by

$$u_y = \sin \phi, \quad \phi = \phi(y, \tau), \quad \left(-\frac{\pi}{2} < \phi < \frac{\pi}{2}, \quad \text{mod } 2\pi\right). \quad (2.7)$$

It follows from (2.5) and (2.7) that

$$\frac{1}{r} = \cos \phi. \quad (2.8)$$

Substituting (2.7) and (2.8) into (2.6), we find

$$\phi_\tau = \sin u. \quad (2.9)$$

Let σ and σ' be solutions of sG equation

$$\sigma_{\tau y} = \sin \sigma, \quad \sigma = \sigma(\tau, y), \quad (2.10)$$

$$\sigma'_{\tau y} = \sin \sigma', \quad \sigma' = \sigma'(\tau, y). \quad (2.11)$$

Then we put

$$u = \frac{1}{2}(\sigma + \sigma'), \quad (2.12)$$

$$\phi = \frac{1}{2}(\sigma - \sigma'). \quad (2.13)$$

In terms of σ and σ' , equations (2.7) and (2.9) can be written as

$$\frac{1}{2}(\sigma + \sigma')_y = \sin \frac{1}{2}(\sigma - \sigma'), \quad (2.14)$$

$$\frac{1}{2}(\sigma - \sigma')_\tau = \sin \frac{1}{2}(\sigma + \sigma'), \quad (2.15)$$

which is a Bäcklund transformation (BT) of the sG equation with a Bäcklund parameter taken to be 1.

Introducing the dependent variable transformation

$$\sigma = 2i \ln \frac{\bar{f}}{f}, \quad (2.16)$$

$$\sigma' = 2i \ln \frac{\bar{g}}{g}, \quad (2.17)$$

where $\bar{f}$ and $\bar{g}$ denote the complex conjugate of f and g , respectively, then we have the bilinear equations

$$D_\tau D_y f \cdot f = \frac{1}{2}(f^2 - \bar{f}^2), \quad (2.18)$$

$$D_\tau D_y g \cdot g = \frac{1}{2}(g^2 - \bar{g}^2), \quad (2.19)$$

where the Hirota's bilinear differential operator $D_y^m D_t^k$ is defined [18] by

$$D_y^m D_t^k a \cdot b \equiv \left(\frac{\partial}{\partial y} - \frac{\partial}{\partial y'}\right)^m \left(\frac{\partial}{\partial t} - \frac{\partial}{\partial t'}\right)^k a(y, t) b(y', t')|_{y'=y, t'=t}.$$

Now from the definition of u and ϕ , we have

$$u = \frac{1}{2}(\sigma + \sigma') = i \ln \frac{\bar{F}}{F}, \quad (2.20)$$

$$\phi = \frac{1}{2}(\sigma - \sigma') = i \ln \frac{\bar{G}}{G}, \quad (2.21)$$

where $F = fg$ and $G = f\bar{g}$.

We give two sets of bilinear equations of 2DTL with tau functions τ_n and τ'_n respectively,

$$\left(\frac{1}{2}D_{x_{-1}}D_{x_1} - 1\right)\tau_n \cdot \tau_n = -\tau_{n+1}\tau_{n-1}, \quad (2.22)$$

$$\left(\frac{1}{2}D_{x_{-1}}D_{x_1} - 1\right)\tau'_n \cdot \tau'_n = -\tau'_{n+1}\tau'_{n-1}. \quad (2.23)$$

It is known that the bilinear equations

$$(D_{x_{-1}} - 1)\tau_n \cdot \tau'_n + \tau_{n+1}\tau'_{n-1} = 0, \quad (2.24)$$

$$(D_{x_1} - 1)\tau_{n+1} \cdot \tau'_n + \tau_n\tau'_{n+1} = 0, \quad (2.25)$$

is one BT between τ_n and τ'_n . The above two-dimensional Toda-lattice (2DTL) equations (2.22)–(2.23) and BT (2.24)–(2.25) have exact solutions in Casorati determinant form with arbitrary parameters. If we put some restrictions on those parameters in solutions, we can obtain

$$\tau_{n+1} \approx \bar{\tau}_n, \quad \tau'_{n+1} \approx \bar{\tau}'_n, \quad (2.26)$$

which means both τ_n and τ'_n are 2-period sequences. Here $\approx$ means two τ functions are equivalent up to a constant multiple and $\bar{\tau}$ denotes complex conjugate of τ .

Proposition 1 *The following Casorati determinants satisfy bilinear equations (2.22)–(2.25)*

$$\tau_n(x_{-1}, x_1) = \begin{vmatrix} \phi_n^{(1)} & \phi_{n+1}^{(1)} & \cdots & \phi_{n+N-1}^{(1)} \\ \phi_n^{(2)} & \phi_{n+1}^{(2)} & \cdots & \phi_{n+N-1}^{(2)} \\ \vdots & \vdots & \ddots & \vdots \\ \phi_n^{(N)} & \phi_{n+1}^{(N)} & \cdots & \phi_{n+N-1}^{(N)} \end{vmatrix}, \quad (2.27)$$

$$\tau'_n(x_{-1}, x_1) = \begin{vmatrix} \psi_n^{(1)} & \psi_{n+1}^{(1)} & \cdots & \psi_{n+N-1}^{(1)} \\ \psi_n^{(2)} & \psi_{n+1}^{(2)} & \cdots & \psi_{n+N-1}^{(2)} \\ \vdots & \vdots & \ddots & \vdots \\ \psi_n^{(N)} & \psi_{n+1}^{(N)} & \cdots & \psi_{n+N-1}^{(N)} \end{vmatrix}, \quad (2.28)$$

where

$$\phi_n^{(i)} = p_i^n e^{\xi_i} + q_i^n e^{\eta_i}, \quad \psi_n^{(i)} = p_i^n (1 - p_i) e^{\xi_i} + q_i^n (1 - q_i) e^{\eta_i}, \quad (2.29)$$

$$\xi_i = p_i x_1 + p_i^{-1} x_{-1} + \xi_{i0}, \quad \eta_i = q_i x_1 + q_i^{-1} x_{-1} + \eta_{i0}. \quad (2.30)$$

Here, p_i, q_i, ξ_i^0 and η_i^0 are the arbitrary parameters that can take either real or complex values.

One can find the proof of this Proposition in [19] for details. Resembling the deduction of solutions to the coupled SP equation and coupled integrable dispersionless equations, we are ready to give N -soliton solution of the generalized sG equation with $\nu = -1$. If p_i and q_i are real parameters, we impose restrictions on the parameters of above τ functions,

$$q_i = -p_i, \quad \eta_{i0} - \xi_{i0} = i\pi/2 + \chi_{i0}. \quad (2.31)$$

If p_i and q_i are complex parameters with χ_{i0} being a real parameter, we put the restrictions

$$q_i = -\bar{p}_i, \quad \eta_{i0} + 2 \arg(q_i) - \xi_{i0} = i\pi/2 + \chi_{i0}. \quad (2.32)$$

These restrictions on parameters guarantee the 2-period reduction (2.26), that is, two neighboring τ functions in both sequences are complex conjugate with a constant factor.

Moreover, by variable transformation $\tau = 2x_1, y = 2x_{-1}$ and substitution $\tau_0 = f, \tau_1 = \bar{f}, \tau'_0 = g, \tau'_1 = \bar{g}$, from (2.22)–(2.23) we have

$$\begin{aligned} D_\tau D_y f \cdot f &= \frac{1}{2}(f^2 - \bar{f}^2), \\ D_\tau D_y \bar{f} \cdot \bar{f} &= \frac{1}{2}(\bar{f}^2 - f^2), \end{aligned} \quad (2.33)$$

and

$$\begin{aligned} D_\tau D_y g \cdot g &= \frac{1}{2}(g^2 - \bar{g}^2), \\ D_\tau D_y \bar{g} \cdot \bar{g} &= \frac{1}{2}(\bar{g}^2 - g^2). \end{aligned} \quad (2.34)$$

which are bilinear equations of the generalized sG equation. From the BT (2.24)–(2.25), we have

$$(2D_\tau - 1)f \cdot g + \bar{f}\bar{g} = 0, \quad (2.35)$$

$$(2D_y - 1)\bar{f} \cdot g + f\bar{g} = 0. \quad (2.36)$$

From equations (2.35)–(2.36), we get

$$\left(\ln \frac{f}{g}\right)_\tau = \frac{1}{2} - \frac{1}{2} \frac{\bar{f}\bar{g}}{fg}, \quad (2.37)$$

$$\left(\ln \frac{\bar{f}}{g}\right)_y = \frac{1}{2} - \frac{1}{2} \frac{f\bar{g}}{\bar{f}g}, \quad (2.38)$$

that lead to

$$\frac{\partial x}{\partial y} = \frac{1}{r} = \cos \phi = \frac{1}{2} \left(\frac{\bar{f}\bar{g}}{f\bar{g}} + \frac{f\bar{g}}{\bar{f}g} \right) = 1 + \left(\ln \frac{g\bar{g}}{f\bar{f}} \right)_y, \quad (2.39)$$

$$\frac{\partial x}{\partial \tau} = \cos u = \frac{1}{2} \left(\frac{\bar{f}\bar{g}}{fg} + \frac{f\bar{g}}{\bar{f}\bar{g}} \right) = 1 + \left(\ln \frac{g\bar{g}}{f\bar{f}} \right)_\tau. \quad (2.40)$$

Thus we obtain the expression for x by tau functions

$$x = y + \tau + \ln \frac{g\bar{g}}{f\bar{f}}. \quad (2.41)$$

Summarizing the above results, the determinant (N -soliton) solution of the generalized sG equation is given by the following theorem.

Theorem 1 *The parametric form for the N -soliton solution of the generalized sG (1.1) is*

$$u = i \ln \frac{\bar{F}}{F},$$

where $F = fg$. f and g are the determinants given by

$$f = \begin{vmatrix} \phi_0^{(1)} & \phi_1^{(1)} & \cdots & \phi_{N-1}^{(1)} \\ \phi_0^{(2)} & \phi_1^{(2)} & \cdots & \phi_{N-1}^{(2)} \\ \vdots & \vdots & \ddots & \vdots \\ \phi_0^{(N)} & \phi_1^{(N)} & \cdots & \phi_{N-1}^{(N)} \end{vmatrix}, \quad (2.42)$$

$$g = \begin{vmatrix} \psi_0^{(1)} & \psi_1^{(1)} & \cdots & \psi_{N-1}^{(1)} \\ \psi_0^{(2)} & \psi_1^{(2)} & \cdots & \psi_{N-1}^{(2)} \\ \vdots & \vdots & \ddots & \vdots \\ \psi_0^{(N)} & \psi_1^{(N)} & \cdots & \psi_{N-1}^{(N)} \end{vmatrix}, \quad (2.43)$$

with

$$\begin{aligned}\phi_n^{(i)}(\tau, y) &= p_i^n e^{\frac{p_i}{2}y + \frac{1}{2p_i}\tau + \eta_{i0}} + (-\bar{p}_i)^n e^{-\frac{\bar{p}_i}{2}y - \frac{1}{2\bar{p}_i}\tau + \xi_{i0}}, \\ \psi_n^{(i)}(\tau, y) &= p_i^n (1 - p_i) e^{\frac{p_i}{2}y + \frac{1}{2p_i}\tau + \eta_{i0}} + (-\bar{p}_i)^n (1 + \bar{p}_i) e^{-\frac{\bar{p}_i}{2}y - \frac{1}{2\bar{p}_i}\tau + \xi_{i0}}, \\ x &= y + \tau + \ln \frac{\bar{g}}{f}, \quad t = \tau,\end{aligned}$$

and η_{i0}, ξ_{i0} satisfy the restriction conditions (2.31)–(2.32).

3 Integrable semi-discretization of the generalized sG equation

3.1 Bäcklund transformations for the generalized sG equation

Based on the above fact, we construct the integrable spatial-discretization of the generalized sG equation. Similar as the integrable discretization of SP equation, we start with two sets of bilinear equations for the semi-discrete 2DTL equations with the same discrete parameter a

$$\left(\frac{1}{a}D_{x-1} - 1\right)\tau_n(k+1) \cdot \tau_n(k) + \tau_{n+1}(k+1)\tau_{n-1}(k) = 0, \quad (3.1)$$

$$\left(\frac{1}{a}D_{x-1} - 1\right)\tau'_n(k+1) \cdot \tau'_n(k) + \tau'_{n+1}(k+1)\tau'_{n-1}(k) = 0, \quad (3.2)$$

which is linked by an another bilinear equations

$$(D_{x-1} - 1)\tau_n(k) \cdot \tau'_n(k) + \tau_{n+1}(k)\tau'_{n-1}(k) = 0. \quad (3.3)$$

Equations (3.1)–(3.2) can be viewed as Bäcklund transformations for the 2DTL equations in the sense that if $\tau_n(k)$ or $\tau'_n(k)$ is a solution to the 2DTL equations (2.22) or (2.23), so is $\tau_n(k+1)$ or $\tau'_n(k+1)$, while (3.3) is the BT linking the solution of (3.1) to (3.2).

Proposition 2 *The bilinear equations (3.1)–(3.3) admit the following determinant solutions*

$$\tau_n(x_{-1}, k) = \begin{vmatrix} \phi_n^{(1)}(k) & \phi_{n+1}^{(1)}(k) & \cdots & \phi_{n+N-1}^{(1)}(k) \\ \phi_n^{(2)}(k) & \phi_{n+1}^{(2)}(k) & \cdots & \phi_{n+N-1}^{(2)}(k) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_n^{(N)}(k) & \phi_{n+1}^{(N)}(k) & \cdots & \phi_{n+N-1}^{(N)}(k) \end{vmatrix}, \quad (3.4)$$

$$\tau'_n(x_{-1}, k) = \begin{vmatrix} \psi_n^{(1)}(k) & \psi_{n+1}^{(1)}(k) & \cdots & \psi_{n+N-1}^{(1)}(k) \\ \psi_n^{(2)}(k) & \psi_{n+1}^{(2)}(k) & \cdots & \psi_{n+N-1}^{(2)}(k) \\ \vdots & \vdots & \ddots & \vdots \\ \psi_n^{(N)}(k) & \psi_{n+1}^{(N)}(k) & \cdots & \psi_{n+N-1}^{(N)}(k) \end{vmatrix}, \quad (3.5)$$

where

$$\phi_n^{(i)}(k) = p_i^n (1 - ap_i)^{-k} e^{\xi_i} + q_i^n (1 - aq_i)^{-k} e^{\eta_i}, \quad (3.6)$$

$$\psi_n^{(i)}(k) = p_i^n (1 - p_i)(1 - ap_i)^{-k} e^{\xi_i} + q_i^n (1 - q_i)(1 - aq_i)^{-k} e^{\eta_i}, \quad (3.7)$$

with

$$\xi_i = p_i^{-1}x_{-1} + \xi_{i0}, \quad \eta_i = q_i^{-1}x_{-1} + \eta_{i0}.$$

Here p_i , q_i , ξ_{i0} and η_{i0} are arbitrary parameters which can take either real or complex values.

The proof is presented in [20].

Applying a 2-reduction condition $q_i = -p_i$, then we could have each of the τ sequence becomes a sequence of period 2, i.e., $\tau_n \rightleftharpoons \tau_{n+2}$, $\tau'_n \rightleftharpoons \tau'_{n+2}$. Furthermore, by choosing particular values in phase constants, we can make τ_n and τ_{n+1} complex conjugate to each other.

By putting $\tau = 2x_{-1}$, $\tau_0(k) = f_k$, $\tau_1(k) = \bar{f}_k$, (3.1) can be converted into

$$\left(\frac{2}{a}D_\tau - 1\right)f_{k+1} \cdot f_k + \bar{f}_{k+1}\bar{f}_k = 0, \quad (3.8)$$

$$\left(\frac{2}{a}D_\tau - 1\right)\bar{f}_{k+1} \cdot \bar{f}_k + f_{k+1}f_k = 0, \quad (3.9)$$

while by putting $\tau'_0(k) = g_k$, $\tau'_1(k) = \bar{g}_k$, (3.2) can be converted into

$$\left(\frac{2}{a}D_\tau - 1\right)g_{k+1} \cdot g_k + \bar{g}_{k+1}\bar{g}_k = 0, \quad (3.10)$$

$$\left(\frac{2}{a}D_\tau - 1\right)\bar{g}_{k+1} \cdot \bar{g}_k + g_{k+1}g_k = 0. \quad (3.11)$$

The bilinear (3.3) is converted into

$$(2D_\tau - 1)f_k \cdot g_k + \bar{f}_k\bar{g}_k = 0, \quad (3.12)$$

$$(2D_\tau - 1)\bar{f}_k \cdot \bar{g}_k + f_kg_k = 0. \quad (3.13)$$

The bilinear equations constitute (3.10)–(3.12) the BT for the generalized sG equation with $\nu = -1$, based on which the semi-discrete generalized sG equations will be deduced in the subsequent two subsections.

3.2 Semi-discrete generalized sG equation

First, we notice that the bilinear equations (3.10)–(3.12) can be rewritten as

$$\frac{2}{a}\left(\ln \frac{f_{k+1}}{f_k}\right)_\tau - 1 = -\frac{\bar{f}_{k+1}\bar{f}_k}{f_{k+1}f_k}, \quad (3.14)$$

$$\frac{2}{a}\left(\ln \frac{\bar{f}_{k+1}}{\bar{f}_k}\right)_\tau - 1 = -\frac{f_{k+1}f_k}{\bar{f}_{k+1}\bar{f}_k}, \quad (3.15)$$

$$\frac{2}{a}\left(\ln \frac{g_{k+1}}{g_k}\right)_\tau - 1 = -\frac{\bar{g}_{k+1}\bar{g}_k}{g_{k+1}g_k}, \quad (3.16)$$

$$\frac{2}{a}\left(\ln \frac{\bar{g}_{k+1}}{\bar{g}_k}\right)_\tau - 1 = -\frac{g_{k+1}g_k}{\bar{g}_{k+1}\bar{g}_k}, \quad (3.17)$$

and

$$2\left(\ln \frac{f_k}{g_k}\right)_\tau - 1 = -\frac{\bar{f}_k \bar{g}_k}{f_k g_k}, \quad (3.18)$$

$$2\left(\ln \frac{\bar{f}_k}{\bar{g}_k}\right)_\tau - 1 = -\frac{f_k g_k}{\bar{f}_k \bar{g}_k}. \quad (3.19)$$

Introducing two intermediate variables

$$\sigma_k(\tau) = 2i \ln \frac{\bar{f}_k(s)}{f_k(s)}, \quad (3.20)$$

$$\sigma'_k(\tau) = 2i \ln \frac{\bar{g}_k(s)}{g_k(s)}, \quad (3.21)$$

one arrives at a pair of semi-discrete sG equation

$$\frac{1}{2a}(\sigma_{k+1} - \sigma_k)_\tau = \sin\left(\frac{\sigma_{k+1} + \sigma_k}{2}\right), \quad (3.22)$$

$$\frac{1}{2a}(\sigma'_{k+1} - \sigma'_k)_\tau = \sin\left(\frac{\sigma'_{k+1} + \sigma'_k}{2}\right). \quad (3.23)$$

Next we introduce dependent variable transformations

$$u_k(\tau) = \frac{1}{2}(\sigma_k + \sigma'_k) = i \ln \frac{\bar{f}_k \bar{g}_k}{f_k g_k}, \quad (3.24)$$

$$\phi_k(\tau) = \frac{1}{2}(\sigma_k - \sigma'_k) = i \ln \frac{\bar{f}_k g_k}{f_k \bar{g}_k}, \quad (3.25)$$

Prior to deriving the semi-discrete generalized sG equation, we present two useful relations.

Proposition 3

$$\frac{d\phi_k}{d\tau} = \sin u_k, \quad (3.26)$$

$$a \sin\left(\frac{\phi_{k+1} + \phi_k}{2}\right) = \sin\left(\frac{u_{k+1} - u_k}{2}\right). \quad (3.27)$$

The proof is given in [Appendix](#). Furthermore, we define a discrete hodograph transformation

$$\delta_k = 2a \cos\left(\frac{\phi_{k+1} + \phi_k}{2}\right), \quad (3.28)$$

which mimics the continuous case

$$\frac{\partial x}{\partial y} = \cos \phi.$$

Then, we have

$$\begin{aligned}
 \frac{d}{d\tau}(u_{k+1} - u_k) &= \frac{1}{2} \frac{d}{d\tau} (\sigma_{k+1} + \sigma'_{k+1} - \sigma_k - \sigma'_k) \\
 &= a \left(\sin \left(\frac{\sigma_{k+1} + \sigma_k}{2} \right) + \sin \left(\frac{\sigma'_{k+1} + \sigma'_k}{2} \right) \right) \\
 &= 2a \sin \left(\frac{\sigma_{k+1} + \sigma'_{k+1} + \sigma_k + \sigma'_k}{4} \right) \cos \left(\frac{\sigma_{k+1} - \sigma'_{k+1} + \sigma_k - \sigma'_k}{4} \right) \quad (3.29) \\
 &= 2a \sin \left(\frac{u_{k+1} + u_k}{2} \right) \cos \left(\frac{\phi_{k+1} + \phi_k}{2} \right) \\
 &= \delta_k \sin \left(\frac{u_{k+1} + u_k}{2} \right).
 \end{aligned}$$

On the other hand, taking the derivative with respect to δ_k results in

$$\begin{aligned}
 \frac{d\delta_k}{d\tau} &= -a \sin \left(\frac{\phi_{k+1} + \phi_k}{2} \right) \frac{d}{d\tau} (\phi_{k+1} + \phi_k) \\
 &= -a \sin \left(\frac{\phi_{k+1} + \phi_k}{2} \right) (\sin u_k + \sin u_{k+1}) \\
 &= -\sin \left(\frac{u_{k+1} - u_k}{2} \right) (\sin u_k + \sin u_{k+1}) \quad (3.30) \\
 &= \frac{1}{2} \cos \left(\frac{3u_{k+1} - u_k}{2} \right) - \frac{1}{2} \cos \left(\frac{3u_k - u_{k+1}}{2} \right) \\
 &= \cos \left(\frac{u_{k+1} - u_k}{2} \right) (\cos u_{k+1} - \cos u_k).
 \end{aligned}$$

Summarizing the above results, we obtain an integrable semi-discrete analogue of the generalized sG equation with $\nu = -1$ together with its N -soliton solution by the following theorem

Theorem 2 A semi-discrete analogue of the generalized sG equation is of the form

$$\frac{d}{d\tau}(u_{k+1} - u_k) = \delta_k \sin \left(\frac{u_{k+1} + u_k}{2} \right), \quad (3.31)$$

$$\frac{d\delta_k}{d\tau} = \cos \left(\frac{u_{k+1} - u_k}{2} \right) (\cos u_{k+1} - \cos u_k), \quad (3.32)$$

where the lattice parameter δ_k is a function depending on (k, t) defined by

$$\delta_k = 2a \cos \left(\frac{\phi_{k+1} + \phi_k}{2} \right). \quad (3.33)$$

Moreover, the N -soliton solution is given by

$$u_k = i \ln \frac{\bar{f}_k \bar{g}_k}{f_k g_k}, \quad \phi_k = i \ln \frac{\bar{f}_k g_k}{f_k \bar{g}_k}, \quad (3.34)$$

where $f_k, g_k, \bar{f}_k$ and $\bar{g}_k$ are tau functions defined by

$$f_k = \tau_0 \left(\frac{\tau}{2}, k \right), \quad \bar{f}_k = \tau_1 \left(\frac{\tau}{2}, k \right), \quad g_k = \tau'_0 \left(\frac{\tau}{2}, k \right), \quad \bar{g}_k = \tau'_1 \left(\frac{\tau}{2}, k \right), \quad (3.35)$$

with

$$\tau_n \left(\frac{\tau}{2}, k \right) = \left| \phi_{n+j-1}^{(i)} \right|_{1 \leq i, j \leq N}, \quad \tau'_n \left(\frac{\tau}{2}, k \right) = \left| \psi_{n+j-1}^{(i)} \right|_{1 \leq i, j \leq N}, \quad (3.36)$$

$$\phi_n^{(i)}(k) = p_i^n (1 - ap_i)^{-k} e^{\frac{1}{2p_i} \tau + \xi_{i0}} + (-\bar{p}_i)^n (1 + a\bar{p}_i)^{-k} e^{-\frac{1}{2\bar{p}_i} \tau + \eta_{i0}},$$

$$\psi_n^{(i)}(k) = p_i^n (1 - p_i)(1 - ap_i)^{-k} e^{\frac{1}{2p_i} \tau + \xi_{i0}} + (-\bar{p}_i)^n (1 + \bar{p}_i)(1 + a\bar{p}_i)^{-k} e^{-\frac{1}{2\bar{p}_i} \tau + \eta_{i0}}.$$

Next, we show that the semi-discrete generalized sG equation converges to the generalized sG equation in the continuous limit. In the discrete case, the dependent variable u is a function of K and τ , meanwhile, it can be viewed as a function of x and t where x is the spatial coordinate at the K th lattice point

$$x = x_0 + \sum_{k=0}^{K-1} \delta_k, \quad t = \tau. \quad (3.37)$$

Note the relation

$$\sin^2 \left(\frac{u_{k+1} - u_k}{2} \right) = a^2 - \frac{\delta_k^2}{4} \quad (3.38)$$

holds due to (3.27) and (3.28). Thus we have

$$\cos \left(\frac{u_{k+1} - u_k}{2} \right) \equiv \sqrt{1 - a^2 + \delta_k^2/4} \sim 1 - \frac{a^2}{2} + \frac{\delta_k^2}{8}, \quad (a \rightarrow 0).$$

In other words,

$$\frac{d\delta_k}{d\tau} = (1 + \varepsilon^2)(\cos u_{k+1} - \cos u_k),$$

where $\varepsilon^2 = \delta_k^2/8 - a^2/2$. Therefore,

$$\frac{\partial x}{\partial \tau} = \frac{\partial x_0}{\partial \tau} + \sum_{k=0}^{K-1} \frac{d\delta_k}{d\tau} = \sum_{k=0}^{K-1} (1 + \varepsilon^2)(\cos u_{k+1} - \cos u_k) \rightarrow \cos u.$$

Thus

$$\partial_\tau = \partial_t + \frac{\partial x}{\partial s} \partial_x \rightarrow \partial_t + \cos u \partial_x. \quad (3.39)$$

Furthermore, we have

$$\begin{aligned} & \frac{d}{d\tau} \frac{u_{k+1} - u_k}{\delta_k} \\ &= \sin \frac{u_{k+1} + u_k}{2} - \frac{u_{k+1} - u_k}{\delta_k^2} (1 + \varepsilon^2)(\cos u_{k+1} - \cos u_k) \\ &= \sin \frac{u_{k+1} + u_k}{2} + 2 \frac{u_{k+1} - u_k}{\delta_k^2} (1 + \varepsilon^2) \sin \left(\frac{u_{k+1} + u_k}{2} \right) \sin \left(\frac{u_{k+1} - u_k}{2} \right). \end{aligned} \quad (3.40)$$

In the continuous limit $a \rightarrow 0$ ($\delta_k \rightarrow 0$), we have

$$\frac{u_{k+1} - u_k}{\delta_k} \rightarrow u_x, \quad \frac{u_{k+1} + u_k}{2} \rightarrow u, \quad \frac{\sin \left(\frac{u_{k+1} - u_k}{2} \right)}{\delta_k/2} \rightarrow u_x.$$

Equation (3.40) converges to

$$(\partial_t + \cos u \partial_x) u_x = (1 + u_x^2) \sin u, \quad (3.41)$$

which is exactly the generalized sG equation (1.1) with $\nu = -1$.

3.3 Another integrable semi-discrete generalized sine-Gordon equation

Following the hodograph transformation in continuous case, we define a discrete hodograph transformation

$$x_k = 2ka + \tau + \ln \frac{\bar{g}_k g_k}{\bar{f}_k f_k}, \quad (3.42)$$

then the nonuniform mesh can be derived as

$$\delta_k = x_{k+1} - x_k = 2a + \ln \frac{\bar{g}_{k+1} g_{k+1} \bar{f}_k f_k}{\bar{f}_{k+1} f_{k+1} \bar{g}_k g_k}. \quad (3.43)$$

Taking the derivative with respect to τ results in

$$\begin{aligned} \frac{d\delta_k}{d\tau} &= \left(\ln \frac{\bar{f}_k}{\bar{g}_k} + \ln \frac{\bar{f}_k}{\bar{g}_k} - \ln \frac{\bar{f}_{k+1}}{\bar{g}_{k+1}} - \ln \frac{f_{k+1}}{g_{k+1}} \right)_\tau \\ &= \frac{1}{2} \left(1 - \frac{f_k g_k}{\bar{f}_k \bar{g}_k} \right) + \frac{1}{2} \left(1 - \frac{\bar{f}_k \bar{g}_k}{f_k g_k} \right) \\ &\quad - \frac{1}{2} \left(1 - \frac{\bar{f}_{k+1} \bar{g}_{k+1}}{f_{k+1} g_{k+1}} \right) - \frac{1}{2} \left(1 - \frac{f_{k+1} g_{k+1}}{\bar{f}_{k+1} \bar{g}_{k+1}} \right) \\ &= \frac{1}{2} \left(\frac{\bar{f}_{k+1} \bar{g}_{k+1}}{f_{k+1} g_{k+1}} + \frac{f_{k+1} g_{k+1}}{\bar{f}_{k+1} \bar{g}_{k+1}} \right) - \frac{1}{2} \left(\frac{f_k g_k}{\bar{f}_k \bar{g}_k} - \frac{\bar{f}_k \bar{g}_k}{f_k g_k} \right) \\ &= \cos(u_{k+1}) - \cos(u_k). \end{aligned} \quad (3.44)$$

$$\begin{aligned} \frac{d}{d\tau}(u_{k+1} - u_k) &= \frac{1}{2} \frac{d}{d\tau} (\sigma_{k+1} + \sigma'_{k+1} - \sigma_k - \sigma'_k) \\ &= a \sin \left(\frac{\sigma_{k+1} + \sigma_k}{2} \right) + a \sin \left(\frac{\sigma'_{k+1} + \sigma'_k}{2} \right) \\ &= 2a \sin \left(\frac{u_{k+1} + u_k}{2} \right) \cos \left(\frac{\phi_{k+1} + \phi_k}{2} \right). \end{aligned} \quad (3.45)$$

Referring two relations proved in previous subsection, we have

$$2a \cos \left(\frac{\phi_{k+1} + \phi_k}{2} \right) = \sqrt{4a^2 - 4 \sin^2 \left(\frac{u_{k+1} - u_k}{2} \right)}, \quad (3.46)$$

which is defined as Δ_k hereafter. Therefore, we can propose an alternative integrable semi-discrete generalized sine-Gordon equation

Theorem 3 *An alternative semi-discrete analogue of the generalized sG equation is of the form*

$$\frac{d}{d\tau}(u_{k+1} - u_k) = \Delta_k \sin \left(\frac{u_{k+1} + u_k}{2} \right), \quad (3.47)$$

$$\frac{d}{d\tau}(x_{k+1} - x_k) = \cos u_{k+1} - \cos u_k, \quad (3.48)$$

where the lattice parameter Δ_k is a function depending on (k, t) defined by

$$\Delta_k = \sqrt{4a^2 - 4 \sin^2 \left(\frac{u_{k+1} - u_k}{2} \right)}. \quad (3.49)$$

The N -soliton solution is given by the same tau functions but different parameterizations. In the continuous limit, we can similarly show that above semi-discrete generalized sG equation converges to the original generalized sG equation.

4 Integrable and non-integrable self-adaptive moving mesh methods

In this section, we propose four self-adaptive moving mesh methods and one Crank-Nicolson scheme for the generalized sG equation. We also compare the numerical results of these five numerical schemes for one kink solution.

4.1 Numerical schemes

Based on two integrable semi-discrete generalized sG equations and a semi-implicit Euler scheme, we propose the following two integrable self-adaptive moving mesh methods.

Scheme 1:

$$p_k^{n+1} = p_k^n + \Delta t \delta_k^n \sin \left(\frac{u_{k+1}^n + u_k^n}{2} \right), \quad (4.1)$$

$$\delta_k^{n+1} = \delta_k^n + \frac{\Delta t}{2} \left(\cos \left(\frac{3u_{k+1}^n - u_k^n}{2} \right) - \cos \left(\frac{3u_k^n - u_{k+1}^n}{2} \right) \right), \quad (4.2)$$

where $p_k = u_{k+1} - u_k$, $\delta_k = x_{k+1} - x_k$. The superscript n represents the numerical value at $t = n\Delta t$.

Scheme 2:

$$p_k^{n+1} = p_k^n + \Delta t \Delta_k^n \sin \left(\frac{u_{k+1}^n + u_k^n}{2} \right), \quad (4.3)$$

$$\delta_k^{n+1} = \delta_k^n + \Delta t \left(\cos u_{k+1}^n - \cos u_k^n \right). \quad (4.4)$$

where

$$\Delta_k^n = \sqrt{4a^2 - 4 \sin^2 \left(\frac{u_{k+1}^n - u_k^n}{2} \right)}.$$

In addition, we have the following non-integrable semi-discrete generalized sG equation

$$\frac{d}{d\tau} (u_{k+1} - u_k) = \delta_k \sin \left(\frac{u_{k+1} + u_k}{2} \right), \quad (4.5)$$

$$\frac{d}{d\tau} (x_{k+1} - x_k) = \cos u_{k+1} - \cos u_k, \quad (4.6)$$

which leads to a non-integrable self-adaptive moving mesh method.

Scheme 3:

$$p_k^{n+1} = p_k^n + \Delta t \delta_k^n \sin\left(\frac{u_{k+1}^n + u_k^n}{2}\right), \quad (4.7)$$

$$\delta_k^{n+1} = \delta_k^n + \Delta t (\cos u_{k+1}^n - \cos u_k^n). \quad (4.8)$$

Furthermore, based on above non-integrable scheme, we propose the following scheme.

Scheme 4:

$$p_k^{n+1} = p_k^n + \Delta t q_k^{n+\frac{1}{2}}, \quad (4.9)$$

$$\delta_k^{n+1} = \delta_k^n + \Delta t r_k^{n+\frac{1}{2}}. \quad (4.10)$$

where

$$q_k^{n+\frac{1}{2}} = \frac{1}{2} \left(\delta_k^n \sin\left(\frac{u_{k+1}^n + u_k^n}{2}\right) + \delta_k^{n+1} \sin\left(\frac{u_{k+1}^{n+1} + u_k^{n+1}}{2}\right) \right), \quad (4.11)$$

$$r_k^{n+\frac{1}{2}} = \frac{1}{2} \left(\cos u_{k+1}^n + \cos u_{k+1}^{n+1} - \cos u_k^n - \cos u_k^{n+1} \right). \quad (4.12)$$

And we take the following approximation to calculate u_k^{n+1}

$$\begin{aligned} u_k^{n+1} &\sim u_k^n + \Delta t \frac{du_k^n}{d\tau} \\ &= u_k^n + \Delta t \left(\frac{du_0^n}{d\tau} + \sum_{i=1}^k \delta_i^n \sin\left(\frac{u_{i+1}^n + u_i^n}{2}\right) \right). \end{aligned}$$

It is known that self-adaptive moving mesh methods can deal with peak waves much better than other methods. To make the comparison we propose the Crank-Nicolson scheme.

Scheme 5 (Crank-Nicolson scheme):

$$p_k^{n+1} = p_k^n + \Delta t \left(\Delta x \sin u_k^{n+\frac{1}{2}} - \frac{\sin u_{k+1}^{n+\frac{1}{2}} - 2 \sin u_k^{n+\frac{1}{2}} + \sin u_{k-1}^{n+\frac{1}{2}}}{\Delta x} \right), \quad (4.13)$$

$$\sin u_k^{n+\frac{1}{2}} = \frac{\sin u_k^n + \sin u_k^{n+1}}{2}. \quad (4.14)$$

4.2 Numerical experiments

The tau functions for the one-soliton solution are

$$f \propto 1 + i e^\theta, \quad g \propto 1 + i s_1 e^\theta, \quad (4.15)$$

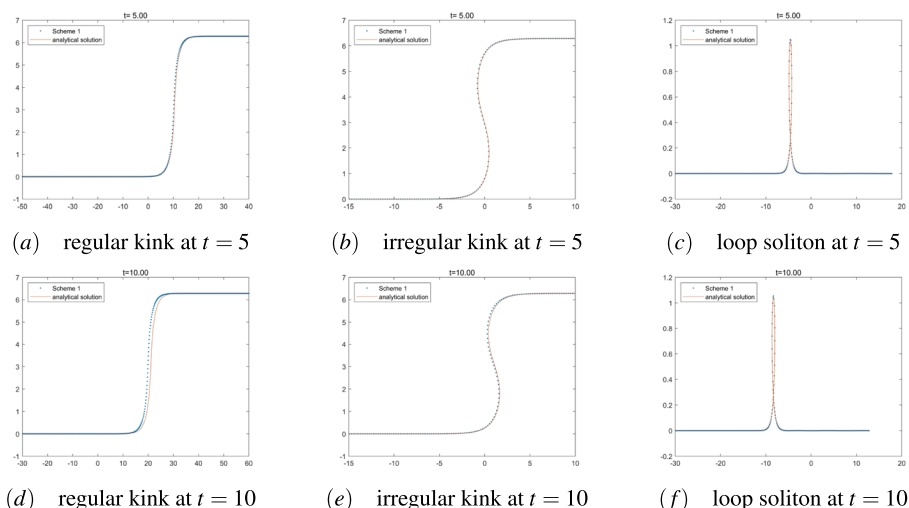


Fig. 1 Comparison between numerical and analytical results for one-soliton solution to the generalized sG equation through the integrable scheme 1. (a) $p_1 = 0.8/\sqrt{2}$, (b) $p_1 = 0.9$, (c) $p_1 = 2$

where $s_1 = \frac{1-p_1}{1+p_1}$, $\theta = p_1 y + \tau/p_1 + \theta_0$. The parametric representation of one-soliton is written in the form

$$u = 2 \tan^{-1}(\sinh \theta - p_1 \cosh \theta) + \pi, \quad (4.16)$$

$$x = y + \tau + \ln \left(\frac{1 + p_1^2}{(1 + p_1)^2} - \frac{2p_1}{(1 + p_1)^2} \tanh \theta \right) + y_0. \quad (4.17)$$

It is analyzed in [3] that one-soliton solution can be classified to different types according to values of p_1 . When $0 < p_1 < \frac{1}{\sqrt{2}}$, it exhibits regular kink solution. When $\frac{1}{\sqrt{2}} < p_1 < 1$, it exhibits irregular kink solution. While $p_1 > 1$, loop soliton appears. We take the initial conditions (4.16)–(4.17) for one-soliton propagation with parameters $y_0 = 0$, and $p_1 = 0.9$, $p_1 = 0.8/\sqrt{2}$ and $p_1 = 2$, respectively. In these three cases, the comparison between the numerical and analytical results by using the integrable scheme 1 is shown in Fig. 1. One can see that the numerical results are in good agreement with the analytical ones. The absolute error of u for one-soliton solutions through scheme 1 is demonstrated in Fig. 2. It shows that the simulation for the

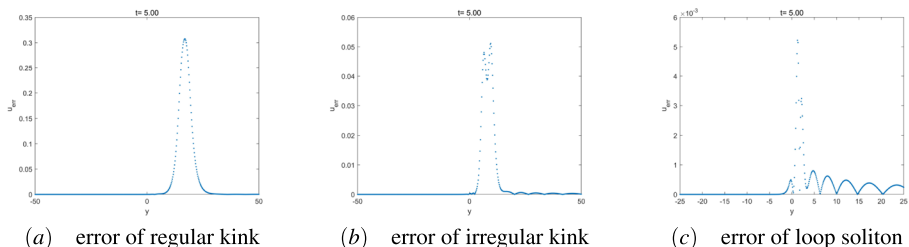


Fig. 2 The absolute error u for one-soliton solution through the integrable scheme 1. (a) $p_1 = 0.8/\sqrt{2}$, (b) $p_1 = 0.9$, (c) $p_1 = 2$

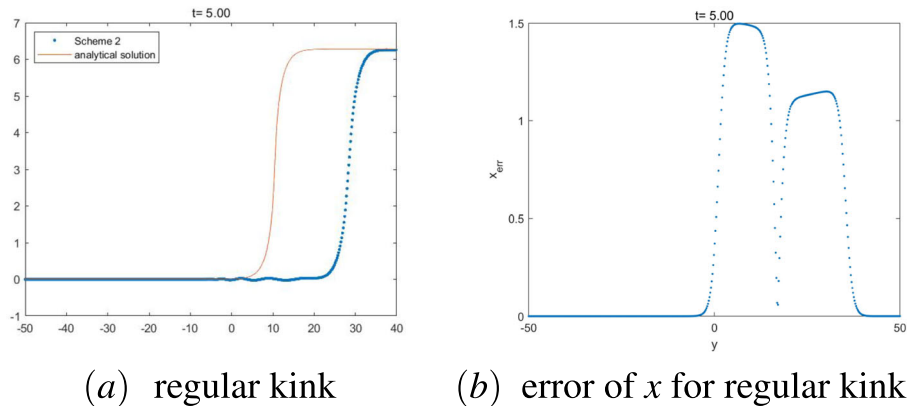


Fig. 3 Numerical experiment for the regular kink solution through the integrable scheme 2 with $p_1 = 0.8/\sqrt{2}$

loop soliton is very accurate. Scheme 2 is also applied to do numerical simulation for the regular kink solution. It is found that the numerical error is much bigger compared with scheme 1 (see Fig. 3(a)). The possible reason is that in scheme 2 the potential p_k is independent with the moving mesh δ_k . The error between analytic and numerical result for x is depicted in Fig. 3(b). Comparison between numerical and analytical results for one-soliton solution to the generalized sG equation through non-integrable scheme 4 is displayed in Fig. 4. Figure 5 shows the comparison among numerical

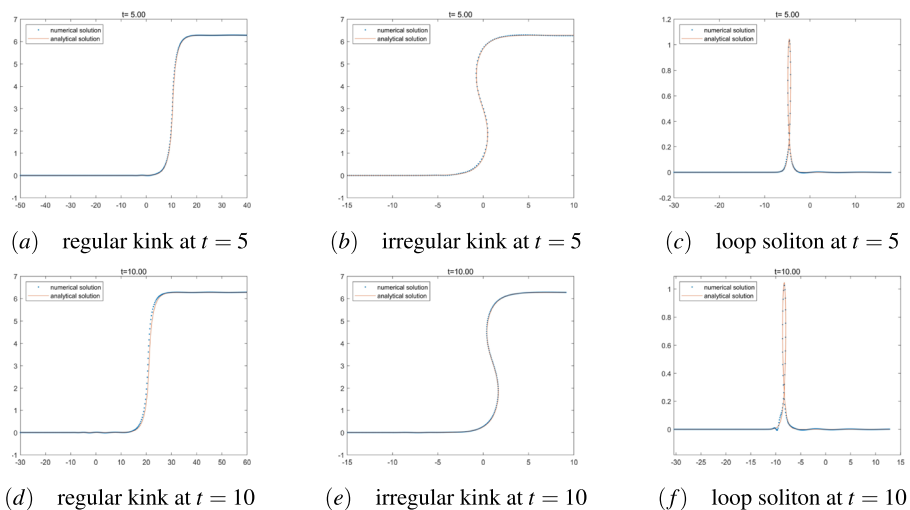


Fig. 4 Comparison between numerical and analytical results for one-soliton solution to the generalized sG equation through scheme 4. (a) $p_1 = 0.8/\sqrt{2}$, (b) $p_1 = 0.9$, (c) $p_1 = 2$

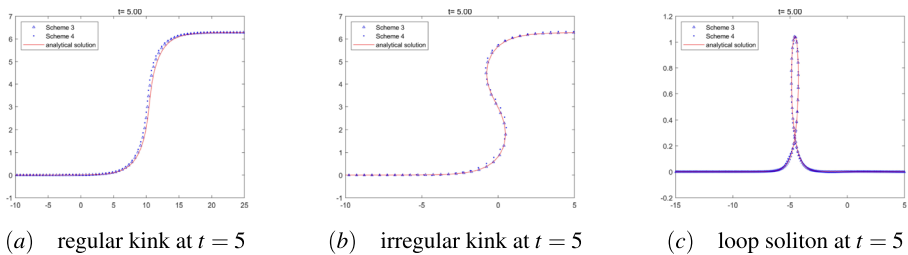


Fig. 5 Comparison between numerical and analytical results for one-soliton solution to the generalized sG equation through scheme 3 and scheme 4. (a) $p_1 = 0.8/\sqrt{2}$, (b) $p_1 = 0.9$, (c) $p_1 = 2$

scheme 3, 4 and analytic solution. The absolute error of u for one-soliton through non-integrable scheme 3 is shown in Fig. 6 and the absolute error of u through non-integrable scheme 4 is displayed in Fig. 7. Compared with the integrable scheme, the simulation for kink and irregular kink solution owns the advantage, while for loop soliton solution, the integrable scheme 1 is much better. Figure 8 exhibits the simulation for regular kink solution by using C-N scheme 5. One can find that the numerical error is much bigger compared with integrable scheme 1 or 2. In particular, the C-N scheme can not simulate the irregular kink and loop soliton because of the multi-value property. From this viewpoint, our self-adaptive moving mesh methods admit much advantage.

5 Conclusion and discussion

In this paper, we proposed two integrable and one non-integrable semi-discrete analogues of a generalized sine-Gordon equation. The determinant formulae of N -soliton solutions for the semi-discrete analogues are presented. The corresponding numerical schemes are given, respectively. The numerical results of one- regular kink, irregular kink and loop soliton are in good agreement with the analytical

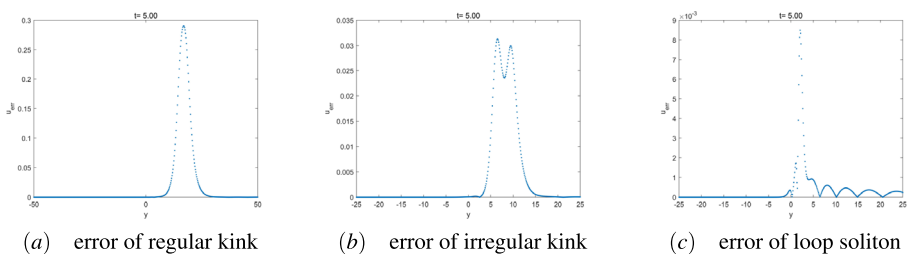


Fig. 6 Absolute error for one-soliton u through the Euler scheme 3. (a) $p_1 = 0.8/\sqrt{2}$, (b) $p_1 = 0.9$, (c) $p_1 = 2$

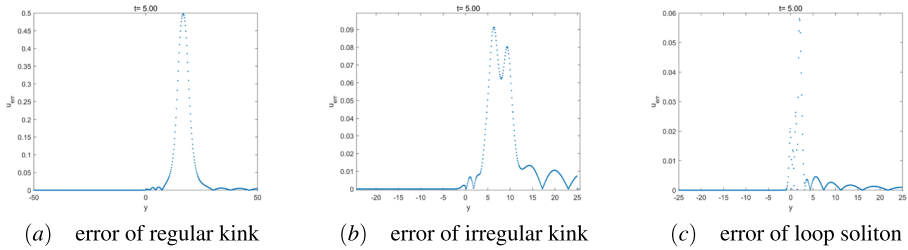


Fig. 7 Comparison between numerical and analytical results for one-soliton solution to the generalized sG equation through the scheme 4. (a) $p_1 = 0.8/\sqrt{2}$, (b) $p_1 = 0.9$, (c) $p_1 = 2$

solution. Moreover, the numerical solutions by integrable self-adaptive moving methods perform better than C-N scheme due to the nature of particle integrability and self-adaptivity of the mesh.

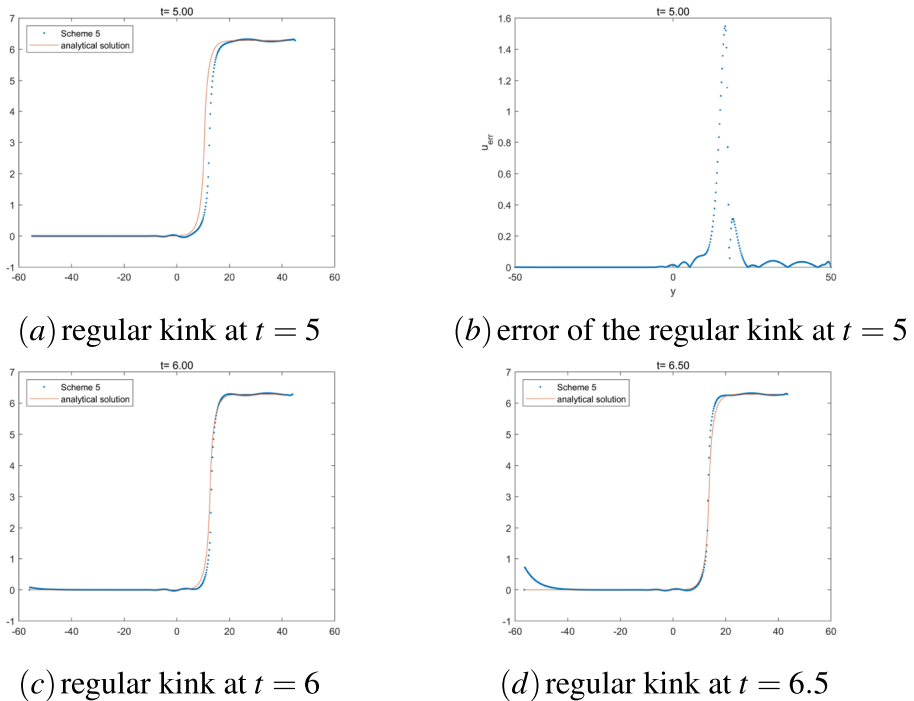


Fig. 8 Comparison between numerical and analytical results for regular kink solution to the generalized sG equation through C-N scheme 5 with $p_1 = 0.8/\sqrt{2}$

Appendix

Here we present the proof of Proposition 3. From the definition of ϕ_k, u_k and equations (3.18)–(3.19), we have

$$\begin{aligned} \frac{d\phi_k}{d\tau} &= i \left(\ln \frac{\bar{f}_k g_k}{f_k \bar{g}_k} \right)_\tau = i \left(\ln \frac{\bar{f}_k}{\bar{g}_k} \right)_\tau - i \left(\ln \frac{f_k}{g_k} \right)_\tau \\ &= \frac{i}{2} \left(\frac{\bar{f}_k \bar{g}_k}{f_k g_k} - \frac{f_k g_k}{\bar{f}_k \bar{g}_k} \right) = \frac{i}{2} \left(e^{-iu_k} - e^{iu_k} \right) \\ &= \sin u_k. \end{aligned} \quad (\text{A.1})$$

Therefore

$$\begin{aligned} \sin u_{k+1} - \sin u_k &= \frac{d}{d\tau} (\phi_{k+1} - \phi_k) \\ &= a \left(\sin \left(\frac{\sigma_{k+1} + \sigma_k}{2} \right) - \sin \left(\frac{\sigma'_{k+1} + \sigma'_k}{2} \right) \right) \\ &= 2a \cos \left(\frac{u_{k+1} + u_k}{2} \right) \sin \left(\frac{\phi_{k+1} + \phi_k}{2} \right), \end{aligned} \quad (\text{A.2})$$

which results in

$$2 \cos \left(\frac{u_{k+1} + u_k}{2} \right) \sin \left(\frac{u_{k+1} - u_k}{2} \right) = 2a \cos \left(\frac{u_{k+1} + u_k}{2} \right) \sin \left(\frac{\phi_{k+1} + \phi_k}{2} \right)?? \quad (\text{A.3})$$

i.e.,

$$a \sin \left(\frac{\phi_{k+1} + \phi_k}{2} \right) = \sin \left(\frac{u_{k+1} - u_k}{2} \right). \quad (\text{A.4})$$

Thus, we complete the proof.

Author contribution B. Feng proposed the original idea and made the review of the manuscript. H. Sheng contributed the numerical simulation. G. Yu conducted the simulation, visualization and wrote the manuscript draft. Both authors read and approved the final manuscript.

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Human and animal ethics Not applicable

Competing interests The authors declare no competing interests.

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